

Module/Pathway Review: Hydroxyproline Catabolism in *Pseudomonas putida* KT2440

Taxon: *Pseudomonas putida* KT2440 (PSEPK, NCBI taxon 160488, proteome UP000000556)

Nominal bucket: KEGG `ppu00470` "D-Amino acid metabolism" (21 candidate genes) **Review**

target: `hydroxyproline_catabolism`

1. Executive summary

The free **trans-4-hydroxy-L-proline (Hyp)** catabolic pathway is present and essentially fully covered in *P. putida* KT2440. It is encoded by a **contiguous four-gene operon, PP_1255–PP_1258**, that converts trans-4-hydroxy-L-proline to 2-oxoglutarate (α -ketoglutarate) in four steps. This is one of the few bacterial lineages where the pathway is biochemically established directly in the target species (Watanabe et al. 2012, [P 22833679](#)),

so module satisfiability rests on strong, direct evidence rather than homology transfer.

The critical curation issue is **not gap-filling but scope**: the assignment inherits the KEGG `ppu00470` "D-amino acid metabolism" bucket, which lumps ~15 unrelated peptidoglycan-, lysine/DAP-, and D-amino-acid-dehydrogenase genes into the candidate list. Only **4 genes (PP_1255, PP_1256, PP_1257, PP_1258)** — plus **paralog disambiguation of the terminal dehydrogenase** — belong to hydroxyproline catabolism. The remaining candidates should be marked `not_expected_in_target_taxon` **for this module** (they are correct genes in *other* modules). Two isomer/derivative pathways that databases frequently conflate — **hydroxyproline betaine** catabolism and **3-hydroxyproline** catabolism — must be kept separate.

Recommended module verdict: all four expected steps `covered`; terminal step additionally flagged `candidate_uncertain` for paralog choice; module boundary `module_needs_revision` (bucket is too broad).

2. Target-organism pathway definition

Process included (and only this): intracellular oxidative catabolism of **free trans-4-hydroxy-L-proline** to the TCA-cycle intermediate **2-oxoglutarate**, allowing Hyp to serve as a carbon/nitrogen source. Trans-4-hydroxy-L-proline is abundant as a collagen breakdown product, and most bacteria cannot use it; *Pseudomonas* is a notable exception

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Bacterial route (distinct from the mammalian D-amino-acid-oxidase route):

1. Epimerization of trans-4-hydroxy-L-proline to cis-4-hydroxy-D-proline.
2. Oxidation of cis-4-hydroxy-D-proline to Δ^1 -pyrroline-4-hydroxy-2-carboxylate (Pyr4H2C).
3. Deamination/ring-opening of Pyr4H2C to α -ketoglutarate semialdehyde (2,5-dioxopentanoate / 2,5-dioxovalerate).
4. Oxidation of α -ketoglutarate semialdehyde to 2-oxoglutarate.

Neighboring pathways / overview maps to keep SEPARATE:

- **KEGG ppu00470 "D-Amino acid metabolism"** (the nominal bucket) — an overview map that also contains peptidoglycan D-Ala/D-Glu metabolism, DAP/lysine biosynthesis, and D-amino-acid dehydrogenases. It is **not** a hydroxyproline module.
- **L-proline degradation** (PutA/proline dehydrogenase → glutamate) — different substrate stereochemistry and enzymes.
- **Hydroxyproline betaine (betonine) / proline betaine (stachydrine) catabolism** — a *betainized* parallel pathway using HpbD-type epimerase/racemase

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Same chemistry class, different substrates and genes; a classic source of annotation cross-contamination.

- **3-hydroxyproline catabolism** (cis-3- / trans-3-Hyp), which runs through an enolase-superfamily dehydratase to Δ^1 -pyrroline-2-carboxylate

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— a different isomer route.

- KEGG maps ppu00040 (pentose/glucuronate) and ppu00300 (lysine) — where several miscataloged candidates actually belong.

Alternate names/IDs: "L-hydroxyproline metabolism," "4-hydroxyproline degradation," "Hyp catabolism"; genes are named *lhp/hyp* in other organisms (e.g., *hypRE/hypO/hypD/hypH* in *Sinorhizobium meliloti*; *lhpABCD* in *Pseudomonas aeruginosa*). MetaCyc: "trans-4-hydroxy-L-proline degradation I."

3. Expected step model

Step	Reaction	EC	Enzyme (family)	KT2440 gene	Status
1	trans-4-hydroxy-L-Pro → cis-4-hydroxy-D-Pro	5.1.1.8	4-Hyp 2-epimerase (HypE; ProR superfamily)	PP_1258 / proR	covered
2	cis-4-hydroxy-D-Pro → Δ^1 -pyrroline-4-hydroxy-2-carboxylate (Pyr4H2C)	1.5.-.- (annot. 1.5.1.-)	D-hydroxyproline dehydrogenase (FAD)	PP_1255	covered
3	Pyr4H2C → α -ketoglutarate semialdehyde (+ NH ₃)	3.5.4.22	Pyr4H2C deaminase (DHDPS/NAL family)	PP_1257	covered
4	α -ketoglutarate semialdehyde → 2-oxoglutarate	1.2.1.26	KG-semialdehyde (2,5-dioxovalerate) dehydrogenase	PP_1256 (operonic)	covered / candidate_uncertain (paralogs PP_2585, PP_3602)

Upstream **transport** of Hyp into the cell is expected but is not part of this enzymatic module and is not represented among the candidate genes (see §5).

4. Candidate genes and evidence

4.1 High-confidence in-pathway genes (the operon PP_1255–PP_1258)

- **PP_1258 / proR (Q88NF3) — 4-hydroxyproline 2-epimerase, EC 5.1.1.8 (Step 1).** Family-level evidence is strong: HypE of the PLP-independent proline-racemase (ProR) superfamily catalyzes the first step of bacterial trans-4-hydroxy-L-proline metabolism (

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Independent confirmation of the identical first step (trans-4-L-Hyp → cis-4-D-Hyp) comes from *S. meliloti* hypRE

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Caveat: ProR-superfamily members can be bifunctional proline racemase / Hyp epimerase, and a single active-site residue (Trp vs Phe) discriminates Hyp from Pro

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Operon context and the 4HypE annotation strongly favor the epimerase function, but the specificity residue is worth verifying at gene-review level.

- **PP_1255 (Q88NF6) — cis-4-hydroxy-D-proline oxidase / D-hydroxyproline dehydrogenase (Step 2). Direct target-species evidence.** Watanabe et al. purified and characterized the *P. putida* enzyme; notably it is a **homomeric, FAD-only** enzyme — i.e., a single gene suffices in KT2440 — in contrast to the heterododecameric ($\alpha_4\beta_4\gamma_4$) *P. aeruginosa* enzyme (

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Curation note: the local EC annotation **1.5.1.-** implies NAD(P) as acceptor, but the enzyme is a **dehydrogenase using artificial/quinone-type electron acceptors** (flavoprotein), so EC should be treated as **1.5.-.-** / **1.5.99.-** -like rather than strictly NAD(P)-linked.

- **PP_1257 (Q88NF4) — Δ^1 -pyrroline-4-hydroxy-2-carboxylate (Pyr4H2C) deaminase, EC 3.5.4.22 (Step 3). Direct target-species evidence** (

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It is a unique member of the dihydrodipicolinate synthase / N-acetylneuraminate lyase

(DHDPS/NAL) (α/β)₈ family, competitively inhibited by pyruvate. The orthologous HypD of *S. meliloti* has a solved structure confirming the NAL-subfamily fold and mechanism (

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- **PP_1256 (Q88NF5) — 2,5-dioxovalerate (α -KG-semialdehyde) dehydrogenase, EC 1.2.1.26 (Step 4).** Best in-pathway candidate on **synteny** (immediately adjacent to PP_1255/57/58). Note it carries a *different* primary bucket in metadata (ppu00040), reflecting that EC 1.2.1.26 is shared with sugar-acid catabolism — an example of how the terminal step is genuinely paralog-ambiguous.

4.2 Terminal-step paralogs (candidate_uncertain)

- **PP_2585 (Q88JR4) and PP_3602 (Q88GW5)** — both annotated EC 1.2.1.26, primary bucket ppu00040. These are broad-specificity aldehyde dehydrogenases likely serving **other** α -ketoglutarate-semialdehyde-producing routes (e.g., oxidative sugar-acid / non-phosphorylative pentose pathways, where α -ketoglutaric semialdehyde dehydrogenase activity is documented in *P. putida*;

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They should not be counted as the hydroxyproline module gene unless expression/knockout data show otherwise.

4.3 Candidates that are over-propagated into this module (belong elsewhere)

All of the following are legitimately annotated but belong to **other** modules; they enter the candidate list only through the broad ppu00470 bucket:

- **Peptidoglycan / cell-wall D-amino-acid:** murI (PP_0736, Glu racemase), murD (PP_1335), ddlA (PP_4346), ddlB (PP_1339).
- **Lysine / diaminopimelate:** lysA-I (PP_2077), lysA-II (PP_5227), dapF (PP_3790), dapF (PP_5228).
- **Alanine / broad racemases:** dadX (PP_5269, Ala racemase), alr (PP_3722, broad-spectrum racemase; characterized as a promiscuous racemase in *Pseudomonas*,

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- **D-amino-acid / D-arginine dehydrogenases:** dadA1 (PP_4434), dadA2 (PP_5270), PP_4311 (DadA small subunit), dauA (PP_2246, catabolic D-Arg dehydrogenase).

- **Amidohydrolase:** ansB (PP_2453, glutaminase-asparaginase).

None of these participates in Hyp → 2-oxoglutarate conversion.

5. Gaps, ambiguities, and likely over-annotations

- **No true enzymatic gap.** All four catalytic steps map to KT2440 genes with direct or strong family evidence. Steps 2 and 3 have direct *P. putida* biochemistry; step 1 has strong family + related-organism evidence; step 4 is present but paralog-ambiguous.
- **Terminal-step paralog ambiguity (main open issue).** Three EC 1.2.1.26 genes (PP_1256, PP_2585, PP_3602); synteny favors PP_1256, but functional assignment of which paralog carries flux for Hyp catabolism is unresolved. → `candidate_uncertain`.
- **EC/annotation precision for PP_1255.** `EC 1.5.1.-` overstates NAD(P)-dependence; the enzyme is a flavin dehydrogenase with artificial acceptors. Curators should relax/correct the acceptor.
- **Transport not represented.** A Hyp uptake system (analogous to *S. meliloti* HypMNPQ ABC transporter;

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is biologically expected but is absent from the candidate list; whether KT2440 uses a dedicated transporter or a general amino-acid permease is an open question (weak transfer from *S. meliloti*).

- **Bucket over-breadth = over-annotation.** ~15 of 21 candidates are unrelated. This is precisely the annotation-propagation problem highlighted for this enzyme class: "at least half of the extant protein annotations are incorrect, and the errors propagate..."

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- **Isomer/derivative confounders.** Keep hydroxyproline *betaine* (HpbD;

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and 3-hydroxyproline (enolase-superfamily dehydratase;

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pathways out of this module.

- **Regulation** (activator/repressor of the PP_1255–1258 operon) is not addressed by the candidate set and remains an open question for KT2440.

6. Module and GO-curation recommendations

Step-level status:

Step	Verdict	Gene(s)
1 Epimerase	covered	PP_1258
2 D-Hyp dehydrogenase	covered	PP_1255
3 Pyr4H2C deaminase	covered	PP_1257
4 KG-semialdehyde dehydrogenase	covered + candidate_uncertain	PP_1256 (assign); PP_2585, PP_3602 (exclude/other-module)
(Transport, optional)	gap / not-in-module	— (no candidate)

Module boundary: `module_needs_revision`. Create/curate a **dedicated hydroxyproline catabolism module** scoped to PP_1255–PP_1258 rather than inheriting KEGG `ppu00470`. Explicitly mark the ~15 peptidoglycan/lysine/D-amino-acid genes as `not_expected_in_target_taxon` **for this module** (they remain valid in their own modules).

GO/EC curation: - PP_1255: prefer GO "D-hydroxyproline dehydrogenase / oxidoreductase acting on CH-NH group of donors, flavin acceptor"; avoid NAD(P)-specific EC `1.5.1.-`. - PP_1257: EC 3.5.4.22 with DHDPS/NAL-family GO; consistent and can stand. - PP_1258: EC 5.1.1.8 (4-Hyp 2-epimerase); flag possible bifunctional proline-racemase activity. - Consider a **GO term request** if no specific term exists for " Δ^1 -pyrroline-4-hydroxy-2-carboxylate deaminase" activity or the assembled "L-hydroxyproline catabolic process."

No new module needed for the core chemistry beyond scoping; separate modules for **hydroxyproline betaine** and **3-hydroxyproline** catabolism should exist independently.

7. Genes to promote to full fetch-gene review

1. **PP_1256** — confirm it (not PP_2585/PP_3602) is the operonic KG-semialdehyde dehydrogenase; resolve the EC 1.2.1.26 paralog assignment. (*highest priority*)
2. **PP_1258 / proR** — verify epimerase vs bifunctional racemase specificity residue (Trp/Phe active-site position).
3. **PP_1255** — correct EC/acceptor annotation (flavin dehydrogenase, not NAD(P)-linked); confirm single-subunit sufficiency.
4. **PP_2585 and PP_3602** — assign to their true pathways (likely sugar-acid / pentose oxidative catabolism) and remove from this module.

The core operon (PP_1255, PP_1257) already has direct *P. putida* biochemistry and needs only annotation tidy-up, not re-investigation.

8. Key references

- P 22833679
— Watanabe et al. (2012) *J Biol Chem*. Identification/characterization of D-hydroxyproline dehydrogenase and Δ^1 -pyrroline-4-hydroxy-2-carboxylate deaminase; **direct *P. putida*/*P. aeruginosa* evidence** for the four-step Hyp $\rightarrow\alpha$ -KG pathway and gene clustering. (*Primary, direct target-species.*)
- P 25786142
— Watanabe et al. (2015). HypE (ProR superfamily) catalyzes the first step (4-Hyp 2-epimerization) of bacterial trans-4-L-Hyp metabolism.
- P 30773259
— Watanabe et al. (2019). Structure of bifunctional proline racemase/Hyp epimerase; Trp/Phe residue governs Hyp vs Pro specificity. (*Specificity caveat for PP_1258.*)
- P 26833407
— Chen et al. (2016) *Sinorhizobium meliloti* L-Hyp/D-proline catabolism: hypRE/hypO/hypD/hypH step assignments, HypD structure, HypMNPQ transporter. (*Related-organism corroboration; moderate transfer.*)

- P** 24520058

— Kumar et al. (2014). Proline betaine / hydroxyproline betaine catabolic pathway (HpbD); explicit warning on annotation over-propagation. (*Boundary separation + over-annotation caution.*)
- P** 25608448

— Zhang et al. (2015). cis-3-hydroxy-L-proline dehydratase (enolase superfamily). (*3-Hyp isomer route to keep separate.*)
- P** 30008699

— Radkov & Moe (2018). Broad-spectrum racemase (Alr) in *Pseudomonas*. (*Reassigns candidate PP_3722/alr out of this module.*)
- P** 19270113

— Meijnen et al. (2009). α -Ketoglutaric semialdehyde dehydrogenase in *P. putida* oxidative sugar catabolism. (*Context for KGSADH paralog function.*)

Evidence grading: Steps 2–3 rest on **direct *P. putida* KT2440-genus biochemistry**; step 1 on **strong family + related-organism** evidence; step 4 presence is certain but **specific-gene assignment is inferential (synteny)**. Transport and regulation are **inferred/unknown** for KT2440. The dominant curation action is scoping the module away from the broad `ppu00470` bucket, not filling enzymatic gaps.
